

which clicked whenever prices changed. So each morning, on the opening, I would see and hear the market click down 200 points, the daily limit, stay unchanged at that price, and know that I had lost that much more, with the amount of potential additional losses still undefined. It was a very tactile experience.

What did you learn from that experience?

It taught me the importance of risk controls because I never wanted to experience that pain again. It enhanced my fear of being wrong and taught me to make sure that no single bet, or even multiple bets, could cause me to lose more than an acceptable amount. In trading you have to be defensive and aggressive at the same time. If you are not aggressive, you are not going to make money, and if you are not defensive, you are not going to keep money. I believe that anyone who has made money in trading has had to experience horrendous pain at some point. Trading is like working with electricity; you can get an electric shock. With that pork belly trade and other trades, I felt the electric shock and the fear that comes with it. That led to my attitude: *Let me show you what I think, and please knock the hell out of it.* I learned about the math of investing.

[Dalio walks over to the board and draws a diagram where the horizontal axis represents the number of investments and the vertical axis the standard deviation.]

This is a chart that I teach people in the firm, which I call *the Holy Grail of investing*.

[He then draws a curve that slopes down from left to right—that is, the greater the number of assets, the lower the standard deviation.]

This chart shows how the volatility of the portfolio changes as you add assets. If you add assets that have a 0.60 correlation to the other assets, the risk will go down by about 15 percent as you add more assets, but that's about it, even if you add a thousand assets. If you run a long-only equity portfolio, you can diversify to a thousand stocks and it will only reduce the risk by about 15 percent, since the average stock has about a 0.60 correlation to another stock. If, however, you're combining assets that have an average of zero correlation, then by the time you diversify to only 15 assets, you can cut the volatility by 80 percent. Therefore, by